Home exam

FIE401

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How does market fragmentation affect stock‘s liquidity?

NHH

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1 Abstract

We study how market fragmentation affects stock liquidity using two regulatory shocks to the trading venue structure of Swiss-listed equities. The lapse of EU equivalence in July 2019 forced trading to consolidate on the SIX Swiss Exchange. The reinstatement of UK equivalence in February 2021 partially restored cross-venue trading. In a difference-in-differences design, we find that the consolidation widened bid-ask spreads for SMI stocks by 2.7 basis points relative to the control group, significant at the 5% level. The 2021 partial re-fragmentation did not reverse the spread increase, suggesting that venue competition matters more for liquidity than single-venue depth.

2 Intro

In this study, we examine how market fragmentation affects stock liquidity. The theoretical prediction is ambiguous: trading on several venues may improve liquidity through competition for order flow, but it may also reduce depth at each individual venue. We use two regulatory shocks to Swiss-listed equities to study this question. On 1 July 2019, the lapse of EU stock market equivalence forced trading in Swiss shares to consolidate on the SIX Swiss Exchange. On 8 February 2021, UK equivalence was reinstated, partly re-fragmenting the market.

We use SMI constituents as the treatment group and CAC40 and DAX30 constituents as the control group. Our empirical strategy is a difference-in-differences design with stock and day fixed effects. The main outcome variable is the quoted bid-ask spread, while Amihud illiquidity is used as a robustness measure. We first document pre-period differences between the groups, especially lower bid-ask spreads for SMI stocks. These level differences are absorbed by stock fixed effects in the regression analysis.

We find that the July 2019 consolidation widened SMI bid-ask spreads by approximately 2.7 basis points relative to the control group. The effect is statistically significant in our preferred specification with controls and is supported by the Amihud illiquidity results. The partial re-fragmentation in 2021 does not reverse the effect, suggesting that the post-2021 market structure did not restore the liquidity conditions from before the 2019 equivalence lapse.

3 Data

We use daily firm-level data from LSEG Workspace for constituents of the Swiss Market Index (SMI), CAC40, and DAX30 from 2 January 2019 to 30 December 2021. Index membership comes from a separate constituents file, and CHF/EUR exchange rates come from the exchange rate file. We merge these sources by firm and date to construct the final firm-day panel. SMI stocks form the treatment group, while CAC40 and DAX30 stocks form the control group.

The sample period is chosen to cover both regulatory events while keeping the analysis close to the shocks of interest. We do not extend the pre-period back to 2018 because a longer window may introduce additional changes in market conditions and firm characteristics that are less directly related to the equivalence events.

We clean the data in the following ways. We remove duplicate firm-day observations and observations with missing or zero trading volume to ensure that liquidity measures are based on valid trading days. We keep stocks with at least 100 trading days and an average price of at least one euro to exclude very thinly traded stocks and potential data errors. We also drop ISIN NL0000235190 that appears in both CAC40 and DAX30, and we multiply DAX30 trading volume by ten to correct the data scaling issue. SMI prices are converted from Swiss francs to euros using the daily CHF/EUR rate. All variables are winsorized at the 0.5% and 99.5% level. These limits reduce the influence of extreme observations while preserving more of the cross-sectional variation than wider filters such as 1%/99% or 5%/95%.

Our main liquidity measure is the quoted bid-ask spread, computed as $(ask - bid)/((ask + bid)/2)$ and reported in basis points. We use the quoted spread as the primary outcome because it directly measures the cost of immediate execution, which is the liquidity dimension most closely linked to venue competition. If competition between venues improves liquidity, this should be reflected in tighter quoted spreads. We do not use turnover as an outcome because trading volume may mechanically reflect the venue structure itself when orders are split across venues. Instead, we keep turnover as a control variable. We also use Amihud illiquidity, calculated as absolute daily return divided by euro trading volume and scaled by 10^9, as a robustness measure because it captures the price-impact dimension of liquidity. As control variables, we use log market capitalisation, annualised turnover, book-to-market, and leverage because firm size, trading activity, valuation, and financial risk are standard determinants of liquidity and may change over time. After filtering, the treatment group consists of 22 SMI constituents with 16,588 firm-day observations, and the control group consists of 85 CAC40 and DAX30 constituents with 61,622 firm-day observations. We divide the sample into a pre-period, a consolidation period, and a re-fragmentation period based on the two regulatory events.

Table 1: Summary Statistics

	Mean	Median	SD	Min	Max
Panel A: SMI (N = 16,588)					
Bid-Ask Spread	3.982	3.324	2.988	0	44.124
log(Mcap)	3.356	3.165	0.975	1.459	5.659
Turnover	0.903	0.721	0.685	0.119	14.42
BTM	0.527	0.349	0.507	0.058	2.892
Leverage (D/E)	1.211	0.6	1.437	0	6.012
Panel B: CAC40/DAX30 (N = 61,622)					
Bid-Ask Spread	21.34	11.585	23.501	0	123.454
log(Mcap)	3.146	3.1	0.991	0.836	5.659
Turnover	0.787	0.325	1.844	0.001	14.42
BTM	0.895	0.536	1.044	0.058	6.598
Leverage (D/E)	1.317	0.778	1.473	0	8.353

This table shows the summary statistics for the variables used in this study. Panel A displays the results for the SMI treatment group, while Panel B provides a summary for the CAC40/DAX30 control group. Bid-Ask Spread is measured in basis points, log(Mcap) is log market capitalisation in billion EUR, Turnover is annualised share turnover, BTM is book-to-market, and Leverage is total debt to common equity. Variables have been winsorized at the 0.5% level.

Table 1 reports summary statistics for the main variables. SMI stocks have substantially narrower spreads on average than the control stocks (3.98 versus 21.34 basis points). The two groups are similar in market capitalisation, turnover, and leverage, while SMI stocks have lower book-to-market values. These differences are considered in the empirical analysis.

4 Analysis

4.1 Pre-period balance

Table 2 presents OLS regressions of each variable on the SMI dummy using only observations before the July 2019 event. Standard errors are clustered at the stock level. SMI stocks have significantly lower bid-ask spreads than the control group in the pre-period, by about 19 basis points. They also have lower book-to-market ratios. For log market capitalisation, turnover, and leverage, the differences are not statistically significant.

Table 2: Pre-period Comparison - SMI vs. CAC40/DAX30

	Bid-Ask Spread	log(Mcap)	Turnover	BTM	Leverage (D/E)
	(1)	(2)	(3)	(4)	(5)
SMI	-18.962^{***}	0.130	0.133	-0.332^{**}	-0.082
	t = -7.710	t = 0.561	t = 0.612	t = -2.323	t = -0.253
Pre-period	Yes	Yes	Yes	Yes	Yes
Cluster SE	By stock	By stock	By stock	By stock	By stock
Observations	12,641	12,641	12,641	12,641	12,641
Adjusted R^2	0.106	0.003	0.001	0.025	0.001
Note:					$^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

This table shows the results of OLS regressions of each variable on an SMI indicator for the pre-event period. The SMI indicator is equal to 1 for SMI constituents and 0 for CAC40/DAX30 constituents. Detailed variable definitions can be found in Table 1. Variables have been winsorized at the 0.5% level. Standard errors are clustered at the stock (ISIN) level.

These differences show that SMI stocks are more liquid already before treatment. This is important for interpretation, but it does not by itself invalidate the difference-in-differences design. Since our main regressions include stock fixed effects, time-invariant level differences between stocks are absorbed. The key identifying assumption is therefore that SMI and control stocks would have followed parallel changes in liquidity absent the consolidation.

4.2 Parallel trends

Figure 1 plots average monthly bid-ask spreads for SMI and the control group. In the pre-period, the two series move similarly. After the 1 July 2019 consolidation, SMI spreads increase relative to the control group. The divergence is visible already in the second half of 2019, before the COVID-19 pandemic. This supports the parallel trends assumption used in the DiD analysis.

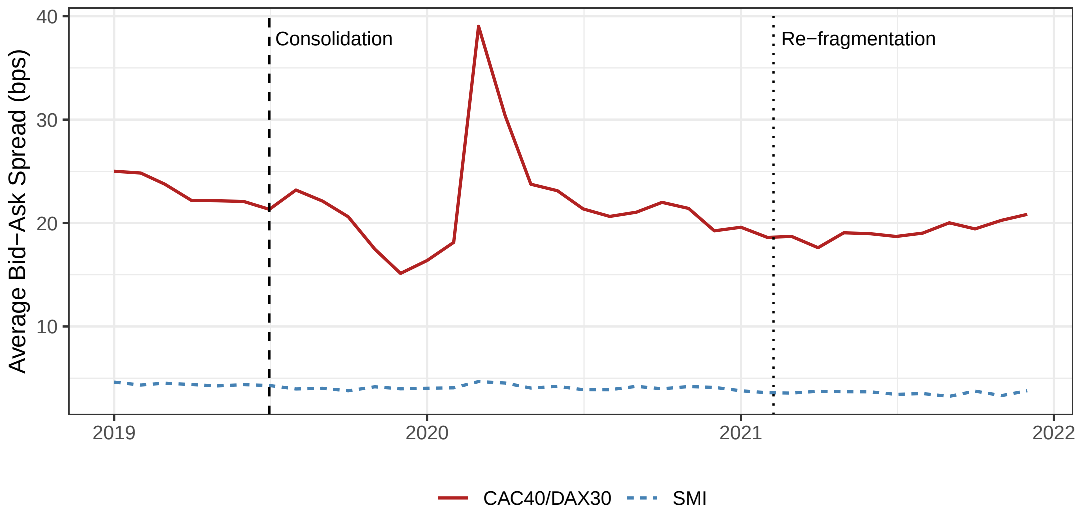

Figure 1: Average Monthly Bid-Ask Spreads

4.3 Main results

Table 3 presents the main difference-in-differences results. Columns 1 and 2 estimate the effect of the July 2019 consolidation on bid-ask spreads. Columns 3 and 4 use the full sample and allow the consolidation and re-fragmentation periods to have separate effects. Columns 5 and 6 use Amihud illiquidity as an alternative liquidity measure. All specifications include stock and day fixed effects, and standard errors are clustered at the stock level.

Table 3: Effect of Market Consolidation and Re-fragmentation on Stock Liquidity

	Dependent variable:					
	Bid-Ask Spread				Amihud Illiquidity	
	(1)	(2)	(3)	(4)	(5)	(6)
SMI x Post	1.582	2.655**			1.227	1.971*
	t = 1.176	t = 1.979			t = 1.361	t = 1.679
SMI x Consolidated			1.555	2.377*		
			t = 1.161	t = 1.785		
SMI x Re-fragmented			3.890**	4.444***		
			t = 2.390	t = 2.663		
log(Mcap)		−9.738***		−7.794***		−14.520*
		t = −3.072		t = −3.276		t = −1.792
Turnover		−1.493***		−1.123***		−0.796**
		t = −3.265		t = −3.269		t = −2.319
BTM		1.766		1.065		−4.919
		t = 0.551		t = 0.389		t = −1.448
Leverage (D/E)		−0.131		−0.226		−2.961*
		t = −0.139		t = −0.262		t = −1.670
Sample	Pre+Cons.	Pre+Cons.	Full	Full	Pre+Cons.	Pre+Cons.
Controls	No	Yes	No	Yes	No	Yes
Observations	54432	54432	78210	78210	54325	54325
Stock FE	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes
Cluster SE	By stock	By stock	By stock	By stock	By stock	By stock
Adjusted R^2	−0.011	0.012	−0.009	0.009	−0.012	−0.001
Note:						$^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

This table shows the results of two-way fixed effects difference-in-differences regressions. SMI × Post is equal to 1 for SMI constituents after 1 July 2019, and 0 otherwise. SMI × Consolidated and SMI × Re-fragmented are equal to 1 for SMI constituents in the consolidation period and post-re-fragmentation period, respectively. Columns (1)–(4) use the quoted bid-ask spread in basis points, while Columns (5)–(6) use Amihud illiquidity. All regressions include stock and day fixed effects.

In our preferred specification with controls (Column 2), the coefficient on SMI × Post is 2.66 and statistically significant at the 5% level. This means that the consolidation increased SMI bid-ask spreads by about 2.7 basis points relative to the control group. Without controls (Column 1), the coefficient is positive but not statistically significant.

The full-sample specification gives a similar interpretation. In Column 4, SMI × Consolidated is positive and significant at the 10% level. The re-fragmentation coefficient is also positive and significant at the 1% level. This means that the 2021 partial re-fragmentation did not reverse the spread increase. The result suggests that the post-2021 market structure did not return to the same liquidity environment as before July 2019.

Columns 5 and 6 repeat the main analysis using Amihud illiquidity. The coefficient is positive and significant at the 10% level in the specification with controls. This supports the main conclusion that consolidation reduced liquidity, although the evidence is weaker than for bid-ask spreads. The control variables also have intuitive signs. Larger firms and stocks with higher turnover tend to have lower spreads, consistent with standard liquidity patterns. Taken together, the consolidation reduced rather than improved liquidity for SMI stocks relative to the control group.

4.4 Unobserved factors at the time of the equivalence lapse

A concern with the DiD interpretation is that other events during the sample period may affect SMI and control stocks differently. Examples include the COVID-19 pandemic, Brexit-related changes in European trading, and sector-specific shocks, since the SMI differs from CAC40 and DAX30 in industry composition. Day fixed effects absorb shocks that affect both groups equally, but they do not remove shocks that affect the groups differently. The most important concern is COVID-19, because it occurs during the consolidation period and directly affects market liquidity.

Three points reduce the COVID concern. First, Figure 1 shows that the spread divergence starts in the second half of 2019, before the pandemic. Second, the COVID spike in March 2020 is visible in both groups, so it is not only an SMI shock. Third, we re-estimate Columns 1 and 2 of Table 3 after excluding February–June 2020. In the specification with controls, the SMI $\times$ Post coefficient increases to 4.03 ($t = 3.02$), significant at the 1% level (untabulated), compared with 2.66 ($t = 1.98$) in the main specification. This suggests that the main result is not driven by COVID. If anything, the COVID months weaken the estimated effect.

4.5 Fundamental differences and the asymmetry between consolidation and re-fragmentation

A second concern is that SMI stocks differ from CAC40 and DAX30 stocks in ways that also affect liquidity. For example, the Swiss index is heavily weighted toward pharmaceuticals and large universal banks, while DAX30 has historically been more industrial and CAC40 more diversified. The SMI is also more concentrated, meaning that liquidity developments in a few large firms can have greater influence on the treatment group. These differences may produce asymmetric responses to shocks such as COVID-19 or changes in global risk sentiment. We account for this by including stock fixed effects, which absorb time-invariant firm differences, and by adding time-varying controls such as market capitalisation, turnover, book-to-market, and leverage. The pre-period balance test in Table 2 also shows that the groups are similar on several important observables. Still, the estimates should be interpreted as relative effects for SMI stocks compared with CAC40 and DAX30 stocks.

The difference between the consolidation and re-fragmentation estimates is also important. The 2021 event did not simply return the market to the pre-2019 situation. Before July 2019, Swiss shares could trade on several EU venues. After February 2021, only UK venues regained access, possibly with additional post-Brexit transaction costs. The re-fragmentation was therefore partial. This provides a plausible explanation for why spreads did not fall back after 2021: the degree of venue competition was still lower than before the equivalence lapse.

5 Conclusion

We examine how market fragmentation affects stock liquidity using two regulatory shocks to Swiss-listed equities. We find that the July 2019 consolidation of SMI trading on the SIX Swiss Exchange increased bid-ask spreads by approximately 2.7 basis points relative to CAC40 and DAX30 controls. The effect is statistically significant in our preferred specification and is economically meaningful given the low pre-event spread level for SMI stocks.

We also examine the partial re-fragmentation in February 2021. This event did not reverse the increase in spreads. A likely explanation is that the 2021 change was not a full return to the pre-2019 market structure, since UK venues regained access while EU venues did not. The results therefore suggest that venue competition is important for liquidity, and that concentrating trading on one venue does not necessarily improve transaction costs.

The interpretation has two main limitations. First, the pre-period is short, although Figure 1 supports the parallel trends assumption. Second, COVID-19 affected market liquidity during the sample period. However, our robustness check excluding February–June 2020 gives a stronger consolidation effect, suggesting that the main result is not driven by the pandemic. Overall, the evidence points to a reduction in liquidity after market consolidation.

AI Declaration

We used generative AI as a writing and coding assistant to improve wording, structure, LaTeX formatting, and R-code debugging.